Written HW5

Logan Tanner
I use Neovim (BTW)

Full HW repository at
github.com/LoganCTanner/School

June 11, 2026

1. Suppose X and Y have the joint density function $f_{X,Y}(x,y) = e^{-(x+y)}$ for $x \geq 0,\ y \geq 0$. Let $W = X + Y$.

   (a) Are X and Y independent? Justify.
   Answer:
   If $f_X(x) \cdot f_Y(y) = f_{X,Y}(x,y)$, then they are independent.

$$
\begin{aligned}
f_X(x) &= e^{-x} \int_0^\infty e^{-y} dy \\
&= e^{-x} \left(-e^{-y} \big|_0^\infty \right) \\
&= e^{-x} \left(0 - (-1) \right) \\
&= e^{-x} \\
f_Y(y) &= e^{-y} \int_0^\infty e^{-x} dx \\
&= e^{-y} \left(-e^{-x} \big|_0^\infty \right) \\
&= e^{-y} \left(0 - (-1) \right) \\
&= e^{-y} \\
f_X(x) \cdot f_Y(y) &= e^{-x} \cdot e^{-y} \\
&= e^{-(x+y)} \\
&= f_{X,Y}(x,y)
\end{aligned}
$$

   $\therefore f_{X,Y}(x,y)$ is independent.

(b) Find the cdf of W.

Answer:

$$\begin{aligned}
F_W(w) &= \int_0^w \int_0^{w-x} e^{-x-y} dy dx \\
&= \int_0^w e^{-x} \left(-e^{-y}|_0^{w-x}\right) dx \\
&= \int_0^w e^{-x} \left(1 - e^{x-w}\right) dx \\
&= \int_0^w e^{-x} - e^{-x+x-w} dx \\
&= \int_0^w e^{-x} - e^{-w} dx \\
&= -e^{-x} - xe^{-w}|_0^w \\
&= -e^{-w} - we^{-w} - (-1 - 0) \\
&= -e^{-w} - we^{-w} + 1
\end{aligned}$$

(c) Use (b) to find the pdf of W.
Answer:

$$\begin{aligned} f_W(w) &= -e^{-w} - we^{-w} + 1dx \\ &= e^{-w} - (e^{-w} + (-we^{-w}) \\ &= e^{-w} + we^{-w} - e^{-w} \\ &= we^{-w} \end{aligned}$$

(d) Now find the pdf of W using the convolution formula (from the beginning of Chapter 6.3).

2. A bin of five transistors is known to contain two that are defective. They are to be tested, one at a time, until the defective ones are identified. Let X be the number of tests needed to find one defective transistor, and let Y be the additional tests needed to find the second. Note that it's possible that a defective transistor can be found even if it wasn't tested, by process of elimination.

(a) Find the joint pmf of X and Y.
Answer:

$$
\begin{aligned}
&P(Y = 0 \cap X = 1) = 0\\
&P(Y = 0 \cap X = 2) = 0\\
&P(Y = 0 \cap X = 3) = \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} = \frac{1}{10}\\
&P(Y = 1 \cap X = 1) = \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{10}\\
&P(Y = 1 \cap X = 2) = \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} = \frac{1}{10}\\
&P(Y = 1 \cap X = 3) = \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3} = \frac{1}{5}\\
&P(Y = 2 \cap X = 1) = \frac{2}{5} \cdot \frac{3}{4} \cdot \frac{1}{3} = \frac{1}{10}\\
&P(Y = 2 \cap X = 2) = 2 \cdot \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} = \frac{1}{5}\\
&P(Y = 2 \cap X = 3) = 0\\
&P(Y = 3 \cap X = 1) = \frac{2}{5} \cdot \frac{3}{4} \cdot \frac{2}{3} = \frac{1}{5}\\
&P(Y = 3 \cap X = 2) = 0\\
&P(Y = 3 \cap X = 3) = 0
\end{aligned}
$$

With the multiple of 2 in $X = 2 \cap Y = 2$ coming from there being two arrangements of GDG{GD,DG}, where G = good and D = defective, and the fractions obtained through $\frac{\text{\# G or D}}{\text{\# Total}}$

(b) Find $p_{Y|X}(y|x)$ for each $x = 1,2,3$.

$$P(Y|1) = \sum_{0}^{3} P(Y = i|1) = \frac{1}{10} + \frac{1}{10} + \frac{1}{5} = \frac{2}{5}$$
$$P(Y|2) = \sum_{0}^{3} P(Y = i|2) = \frac{1}{10} + \frac{1}{5} = \frac{3}{10}$$
$$P(Y|3) = \sum_{0}^{3} P(Y = i|3) = \frac{1}{10} + \frac{1}{5} = \frac{3}{10}$$

(c) Find $E[Y|X = x]$, for each of $x = 1,2,3$.

$$\begin{aligned}
E[Y|X = 1] &= \sum_{0}^{3} E[Y_i|X = 1] = \sum_{0}^{3} Y_i \cdot P(Y_i|X = 1) \\
&= 0 \cdot 0 + 1 \cdot \frac{1}{10} + 2 \cdot \frac{1}{10} + 3 \cdot \frac{1}{5} \\
&= \frac{1}{10} + \frac{2}{10} + \frac{6}{10} = \frac{9}{10} \\
E[Y|X = 2] &= \sum_{0}^{3} E[Y_i|X = 2] = \sum_{0}^{3} Y_i \cdot P(Y_i|X = 1) \\
&= 0 \cdot 0 + 1 \cdot \frac{1}{10} + 2 \cdot \frac{1}{5} + 3 \cdot 0 \\
&= \frac{1}{10} + \frac{4}{10} = \frac{1}{2} \\
E[Y|X = 3] &= \sum_{0}^{3} E[Y_i|X = 3] = \sum_{0}^{3} Y_i \cdot P(Y_i|X = 1) \\
&= 0 \cdot \frac{1}{10} + 1 \cdot \frac{1}{5} + 2 \cdot 0 + 3 \cdot 0 \\
&= \frac{1}{5}
\end{aligned}$$

3. Let X and Y be independent $unifrom[0,1]$ random variables.

   (a) Find pdf of $|X - Y|$.
   Answer:
   With the marginal pdfs of X and Y both being $f = \frac{1}{1-0} = 1$, and their same intervals over [0,1], it follows that the cdf of $|X - Y|$ would then be

$$\begin{aligned}
P(|X-Y| \le a) &= P(-a \le X - Y \le a) \\
&= 2P(X - Y \le a) \\
&= 2P(-Y \le a - X) \\
&= 2P(Y \ge X - a) \\
&= 2(1 - P(Y \le X - a)) \\
&= 2\left(1 - \int_a^1 \int_0^{x-a} dydx\right) \\
&= 2\left(1 - \int_a^1 x - adx\right) \\
&= 2\left(1 - \left(\frac{1}{2}x^2 - ax\Big|_a^1\right)\right) \\
&= 2\left(1 - \left(\frac{1}{2} - a - \left(\frac{1}{2}a^2 - a^2\right)\right)\right) \\
&= 2\left(1 - \left(\frac{1}{2} - a + \frac{1}{2}a^2\right)\right) \\
&= 2\left(-\frac{1}{2} + a - \frac{1}{2}a^2\right) \\
&= -1 + 2a - a^2
\end{aligned}$$

   and the pdf is then obtained from the derivative of the cdf

$$\begin{aligned}
f_{|X-Y|}(a) &= (-1 + 2a - a^2)da \\
&= \boxed{2 - 2a}
\end{aligned}$$

   (b) Compute $E[|X - Y|]$.
   Answer:
   Given the pdf of $-2 + 2a$, the expected value would then be

$$\begin{aligned}
E[f_{|X-Y|(a)}] &= \int_0^1 a(2 - 2a)da \\
&= a^2 - \frac{2}{3}a^3\Big|_0^1 \\
&= 1 - \frac{2}{3} = \boxed{\tfrac{1}{3}}
\end{aligned}$$

4. Consider an Elevator in a building with 10 floors above the ground floor. The number of people entering the elevator on the ground floor is a Poisson random variable with mean 5, and each person is equally likely to get off at any other floor, independently of the others. What is the expected number of stops the elevator will make to allow all of the passengers to get off?
Answer:
Letting $X = \#$ people on the elevator, and $Y = \#$ stops, then the expected number of stops is given by

$$E[Y] = \sum_{n=0}^{\infty} E[Y|X = n] \cdot P(X = n)$$

where $P(X = n) = e^{-5}\frac{5^n}{n!}$, and $E[Y|X = n] = \sum_1^{10} 1 - \left(\frac{9}{10}\right)^n = 10(1 - \frac{9}{10}^n)$. Then, multiplying the two and separating the sums,

$$\sum_{n=0}^{\infty} 10e^{-5}\frac{5^n}{n!} - \left(\sum_{n=0}^{\infty} 10\left(\frac{9}{10}\right)^n e^{-5}\frac{5^n}{n!}\right)$$

The first sum then sums to 10×1, since the sum is over the poisson which equals 1. The second sum has a similar result, by rearranging to be

$$\sum_{n=0}^{\infty} 10e^{-5}\left(\frac{9 \times 5}{10}\right)^n \cdot \frac{1}{n!} = 10e^{-5}\sum_{n=0}^{\infty}\left(\frac{9 \times 5}{10}\right)^n \cdot \frac{1}{n!}$$

Which then follows the pattern for e^λ, where

$$\sum_{n=0}^{\infty}\frac{\lambda^n}{n!} = e^\lambda \Rightarrow \sum_{n=0}^{\infty}\left(\frac{45}{10}\right)^n \cdot \frac{1}{n!} = e^{4.5}$$

Resulting in

$$10 - 10e^{4.5-5} = 10 - \frac{10}{\sqrt{e}} \approx 3.9347$$

5. Suppose I wrote the lengthy description for this problem,

(a) What do you think might tend to happen to restaurants that have bad food and a bad ambiance? Explain why it might be more reasonable to assume that your friend was only sampling data from part of this domain that lies on or above the line $y = 5 - x$, say.
Answer:
Restaurants that don't perform well will tend to go out of business, so only those that are of the highest quality in at least one of tastiness or ambience will survive long enough to stay open.

(b) Use this updated assumption and new, triangular domain to write a joint density function for X and Y, the tastiness and ambiance scores respectively of a randomly chosen restaurant with at least 10 ratings on Zelp. Also find the marginal densities for X and Y.
Answer:
Since this triangular region is uniformly distributed,

$$f_{X,Y}(x,y) = \frac{1}{\text{Area}} = \frac{1}{\frac{25}{2}} = \frac{2}{25}$$

$$\begin{aligned} f_X(x) &= \frac{2}{25}\int_{5-x}^{5} dy = \frac{2}{25}x \\ f_Y(y) &= \frac{2}{25}\int_{5-y}^{5} dx = \frac{2}{25}y \end{aligned}$$

(c) Find the expected tastiness score of one of these restaurants.
Answer:

$$\begin{aligned} E[X] &= \int_0^5 xf(x,y)dx \\ &= \int_0^5 \frac{2}{25}xdx \\ &= \frac{1}{25}\left(x^2\Big|_0^5\right) \\ &= 1 \end{aligned}$$

(d) Now, using appropriate conditional densities, find the expected tastiness scores of a restaurant with an ambiance score of 5, and a restaurant with an ambiance score of 1.
Answer:

$$
\begin{aligned}
E[X|Y=1] &= \int_0^5 x\frac{f(x,5)}{f_Y(1)}dx \\
&= \int_0^5 x\frac{\frac{2}{25}}{\frac{2}{25}}dx \\
&= \int_0^5 xdx \\
&= \frac{1}{2}(25-0) = \frac{25}{2} \\
E[X|Y=5] &= \int_0^5 x\frac{f(x,5)}{f_Y(5)}dx \\
&= \int_0^5 x\frac{\frac{2}{25}}{\frac{2}{5}}dx \\
&= \frac{1}{5}\int_0^5 xdx \\
&= \frac{1}{10}(25-0) = \frac{25}{10}
\end{aligned}
$$

I'm not sure if I did something wrong here, it feels like this should be at most $\frac{10}{2}$ since the score caps out at 5.

(e) Use your work to give an alternate interpretation of your friend's findings.
Answer:
It is harder to keep a restaurant open with larger expenses, and those with lower ambiance have lower expenses, so there will be a wider range of tastiness scores, whereas those with high ambiance must have high tastiness so they're able to get enough customers to afford to make enough money to stay open.